# Lab Report: Image classification experiments

## 1 Introduction

In this report we compare several models on standard benchmarks. We trained each model for 20 epochs with a batch size of 32. The dataset was split into 70 percent training data and the rest for testing.

## 2 Results

Figure 1 shows the accuracy of Baseline by dataset.

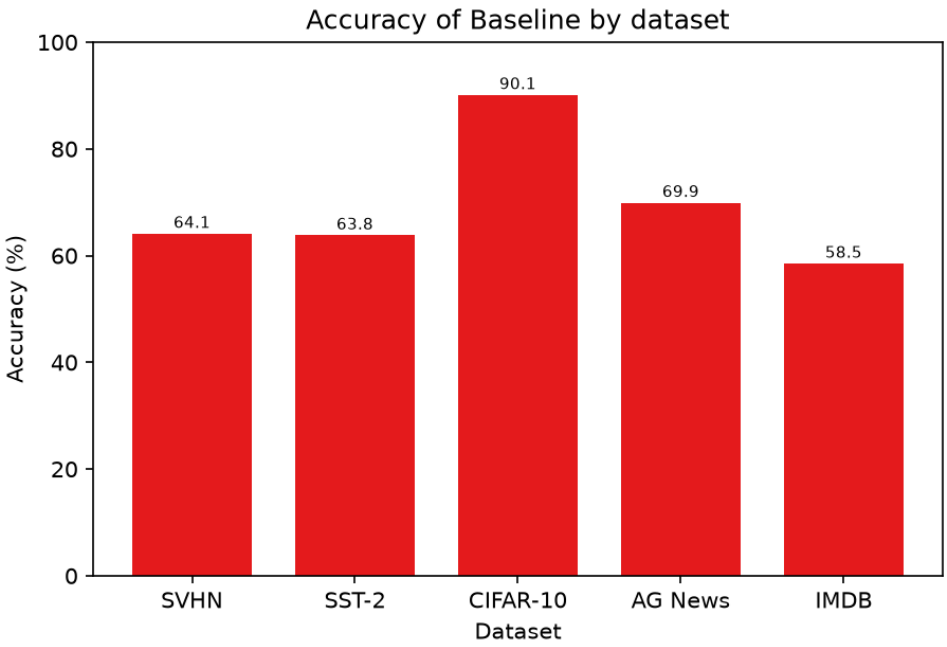



Figure 1: Accuracy of Baseline by dataset for the models in this experiment.

Baseline scores 9.4 points higher on AG News than on IMDB (Figure 1). Baseline posts 58.5% accuracy when tested on IMDB. Evaluated on CIFAR-10, Baseline comes in at 92.1% accuracy.

Figure 2 shows the validation accuracy during training.

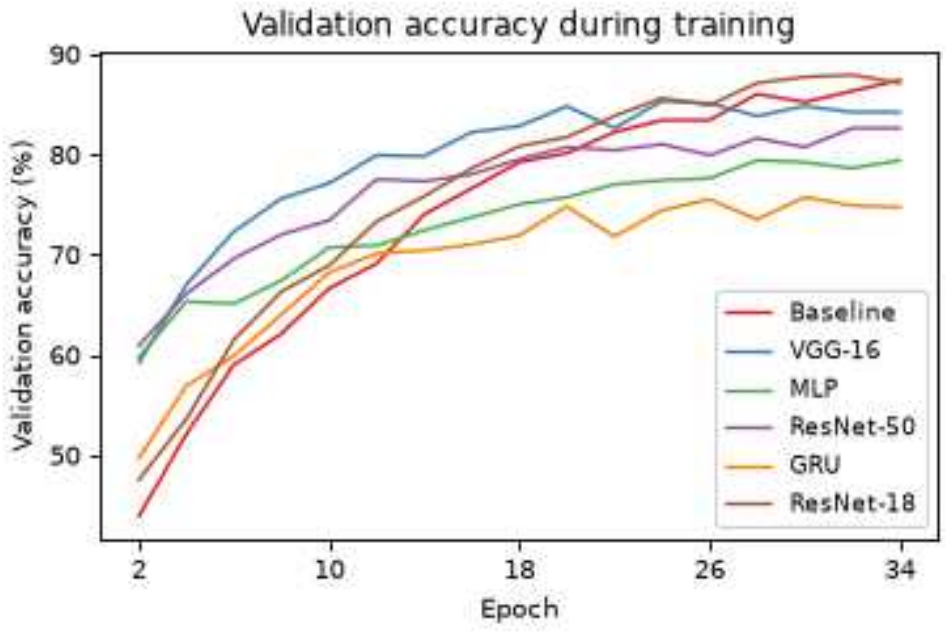



Figure 2: Validation accuracy during training for the models in this experiment.

Epoch 14 sees MLP post a validation accuracy of 77.2%. The validation accuracy of Baseline improves by a

relative 16.0% between epoch 12 and epoch 18. ResNet-18 outperforms Baseline by 6.4 percentage points at epoch 18.

## 3 Discussion

The differences between the models are discussed in the next section.